Framework Introduce

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Vision Tools Script	1.Vision Pro 脚本的编写 & 脚本调试		1	胡建鹏
	2. 工具的调用及其参数的输入输出	工具的调用及其参数的输入输出		
	3.VPP 界面显示（Inspection 输入输出的介绍）	VPP 界面显示（Inspection 输入输出）		
	4.Vision Tools Script Demo 1	练习 Vision Tools Script	1	甲文
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Framework	1.Framework 框架介绍	Framework 框架介绍	1	锦涛
	2. 创建一个新的 MachineSupport	项目启动，创建一个空的 machinesupport。		
	3. 相机调用	相机的设置调用等		
	4. 标定及 inspection 模块	标定及 Inspection 的添加，功能设置等		
	5.IMachineSupport 接口主要函数介绍	MS 中最重要的，要用户实现的接口介绍	1	建军
	6. 其他 Framework 中主要函数介绍	其他常用的 Framework 函数讲解		
	7. 通讯模块	控制 inspection 输入输出	1	认认
		控制对应 Inspection 运行		
		与 SI 通讯的建立，按照 SI 指令运行 inspection，获取结果，发送坐标等结果到 SI。		
	8.Vpp 间数据传输介绍	跨 vpp 数据分享及传递		
	9.MachineSupport 自定义窗口的添加	项目自定义窗体的添加和设计。		
	10. 存图模块	项目存图逻辑如 Group 模块。		
	11. 从 0 开始搭建一个简单的三相机对位项目。	综合演练	1	黄海

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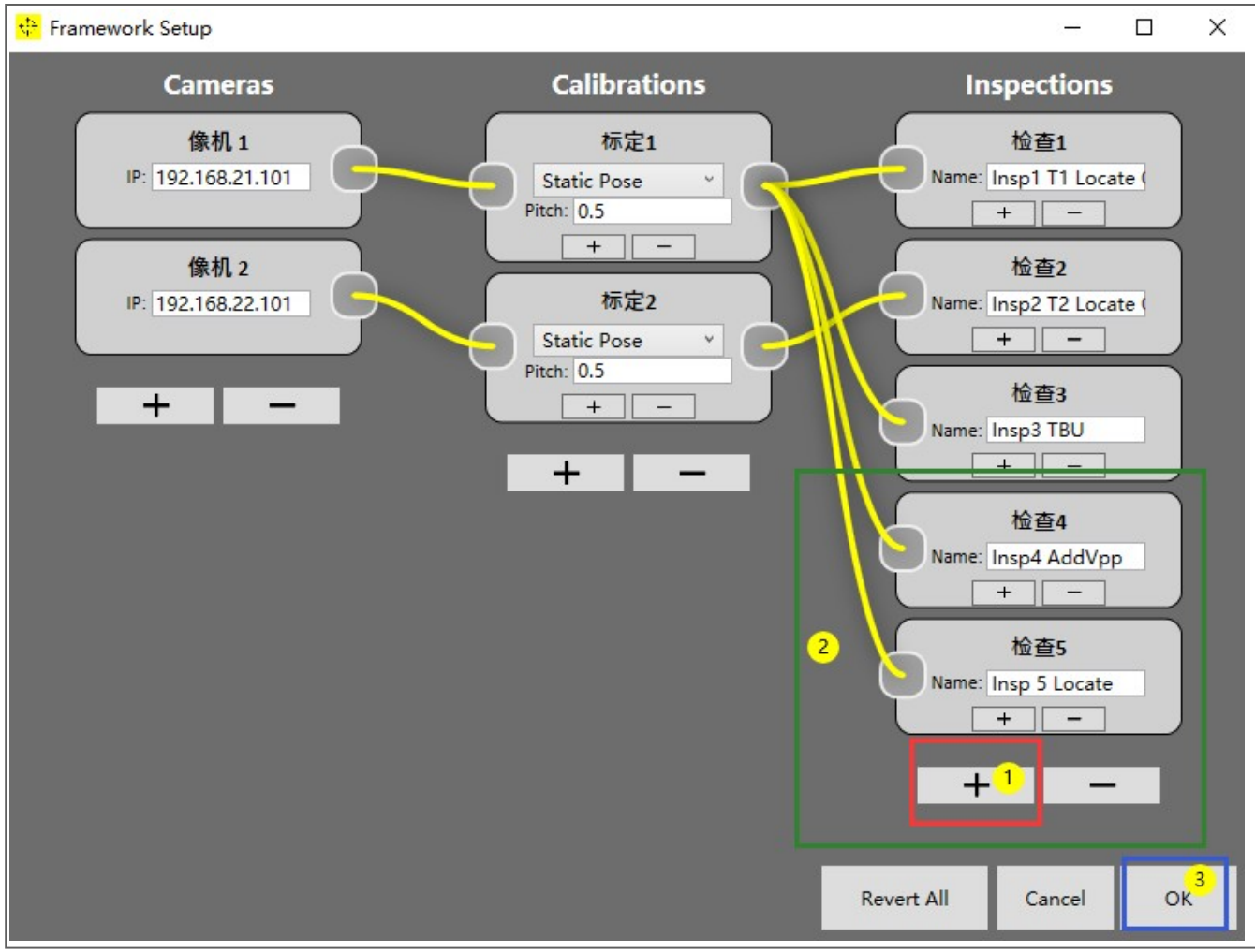
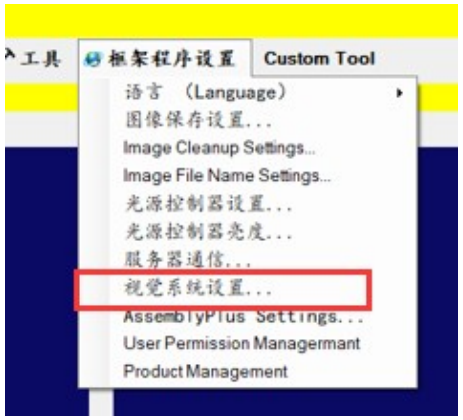
Framework 框架介绍





1. 打开程序后找到：视觉系统设置

2. 按照左图中①②③顺序添加 VPP 并命名连线，最后确定，该名字就是在 Inspection 中显示的名字

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Framework 框架介绍

Framework 文件分布如下：

Calibrations	1/4/2020 5:42 PM	File folder		标定文件夹
Cameras	1/4/2020 5:42 PM	File folder		相机文件夹
CameraSetupTools	1/4/2020 5:42 PM	File folder		相机 Setup 工具文件夹
CamSetup	1/4/2020 5:42 PM	File folder		相机 Setup 文件夹
DisplayColorMaps	1/4/2020 5:42 PM	File folder		显示文件夹
Inspections	1/6/2020 7:29 PM	File folder		Inspection 文件夹
Language	1/4/2020 5:42 PM	File folder		语言文件夹
Log	7/27/2020 11:45 AM	File folder		Log 文件夹
Plugins	1/4/2020 5:42 PM	File folder		插件文件夹
Universal	1/4/2020 5:42 PM	File folder		通用文件夹
CognexAssemblyPlus.exe	12/6/2019 4:26 PM	Application	1,562 KB	启动程序的 exe
coggsgarbo.dll	4/11/2019 4:29 PM	Application extension	1,384 KB	
Cognex.VisionPro.AlignPlusCalibration.dll	10/18/2019 6:35 AM	Application extension	15,079 KB	
Cognex.VisionPro.dll	10/8/2019 5:44 PM	Application extension	298 KB	
Cognex.VisionPro.Vs.TexturedRangeImage.dll	10/18/2019 6:36 AM	Application extension	6,080 KB	
Cognex.VisionPro3D.Core.dll	10/8/2019 5:45 PM	Application extension	42,680 KB	
Cognex.VisionPro3D.dll	10/8/2019 5:45 PM	Application extension	48,103 KB	
Cognex.VPE.AOI.dll	11/19/2019 6:57 AM	Application extension	25 KB	引用相关的 dll
Cognex.VPE.AOIGradingUtility.dll	2/21/2019 2:17 AM	Application extension	113 KB	
Cognex.VPE.AOIShadowCamera.dll	12/6/2019 3:51 PM	Application extension	61 KB	
Cognex.VPE.Calib.dll	2/27/2018 9:45 PM	Application extension	34 KB	
Cognex.VPE.Cami.dll	11/19/2019 6:40 AM	Application extension	13,133 KB	
Cognex.VPE.CGSnap.dll	11/19/2019 6:57 AM	Application extension	11 KB	
Cognex.VPE.Core.dll	11/19/2019 6:36 AM	Application extension	14,522 KB	
Cognex.VS.MachineSupport.dll	2023/1/4 17:08	应用程序扩展	44 KB	MachineSupport 生成类库文件
Cognex.VS.MachineSupport.pdb	2023/1/4 17:08	Program Debug...	84 KB	

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Machine Support

Init

public void Init(IFrameworkSupport framework)

功能：初始化 Framework 等

参数： IFrameworkSupport

ForceRecordFlush

public void ForceRecordFlush()

功能： 记录 InspectionRecordLogger 结果到 CSV ，此功能暂时不用

参数：无参数

MachineName

public string MachineName

功能：更改此函数可以在初始化时 Log 日志里面可以看到此 Fixture 的名称

参数：无参数，只读属性

TCPJobParser

public string TCPJobParser(CommandAndInfo command)

功能：此函数负责分析传入的指令，选择要运行的工具 / 设置输入 / 工具的运行 / 输出的收集 / 形成响应字符串

参数： CommandAndInfo 传入指令信息

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Machine Support

TCPAcqArmed

```
public string TCPAcqArmed(CommandAndInfo command)
```

功能： 收到上位机指令 Framework 准备好一切硬触发工作后通过此命令反馈给上位机信号

参数： CommandAndInfo 传入指令信息

TCPAcqComplete

```
public string TCPAcqComplete(CommandAndInfo command)
```

功能： 取相完成指令

参数： CommandAndInfo 传入指令信息

SimulatorCommandStrings

```
public List<string> SimulatorCommandStrings
```

功能：将模拟指令集增加到界面上

参数：只读属性

SimulatorCommandParserIDs

```
public List<int> SimulatorCommandParserIDs
```

功能： 模拟不同的服务器发来不同的指令，一般为空

参数：只读属性

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Machine Support

ProfinetJobParser

```
public ProfinetUserData ProfinetJobParser(CommandAndInfo command)
```

功能：此函数负责分析传入的指令，和 PLC 通讯相关
参数： CommandAndInfo 传入指令信息

SimulatorPLCCommands

```
public List<int> SimulatorCommandParserIDs
```

功能：将模拟指令集增加到界面上，和 PLC 通讯相关
参数：只读属性

Shutdown

```
public void Shutdown()
```

功能： 程序关闭时将调用此函数来清理 Init 时加载的任何内容
参数：无参数

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Machine Support

```
public string MachineName...
public List<int> SimulatorCommandParserIDs...
public List<string> SimulatorCommandStrings...
public List<Tuple<String, Int16>> SimulatorPLCCommands...

public void ForceRecordFlush()...
public Cognex.VisionPro.CogGraphicCollection GetAcquisitionSetupOverlayGraphics(int
public void Init(IFrameworkSupport framework)...
public ProfinetUserData ProfinetJobParser(CommandAndInfo command)...
public void Shutdown()...
public string TCPAcqArmed(CommandAndInfo command)...
public string TCPAcqComplete(CommandAndInfo command)...
public string TCPJobParser(CommandAndInfo command)...
```

- 返回设备名称
- 返回模拟器数量
- 返回模拟器指令表（String）
- 返回模拟器 PLC 指令表（16 位数字）
- 保存数据
- 实时模式添加 Graphic
- 初始化
- 处理 Profinet 接口
- 关闭
- 收到指令被调用
- 取像完成被调用
- 取像前被调用

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Machine Support

```
namespace Cognex.VS.MachineSupport
{
    public class MachineSupportFactory
    {
        public static IMachineSupport Create()...
    }

    public class MachineSupportSample : IMachineSupport
    {
        private IFrameworkSupport m_framework;

        public string MachineName...
        public List<int> SimulatorCommandParserIDs...
        public List<string> SimulatorCommandStrings...
        public List<Tuple<String, Int16>> SimulatorPLCCommands...

        public void ForceRecordFlush()...
        public Cognex.VisionPro.CogGraphicCollection GetAcquisitionSetupOverlayGraphics(int
        public void Init(IFrameworkSupport framework)...
        public ProfinetUserData ProfinetJobParser(CommandAndInfo command)...
        public void Shutdown()...
        public string TCPAcqArmed(CommandAndInfo command)...
        public string TCPAcqComplete(CommandAndInfo command)...
        public string TCPJobParser(CommandAndInfo command)...
    }
}
```


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Machine Support： 光源的调用

```
runJobParams.Inputs["WedingPadSort"] = padsort;
command.WaitForDisplayResultCompleted = true;
success = m_framework.RunJob(command, LightControlAction.AutoOnAndOff, true, runJobParams);

//20180728/////////////////////////////////////////////////
bRunStatus = success;
/////////////////////////////////////////////////////////
if (!success)
{
    MessageManager.gOnly.Info("Vision Tool Run Error,\r\n
    CogGraphicLabel errorLabel = new CogGraphicLabel();
```

AutoOff
AutoOn
AutoOnAndOff
AutoOnAndOffCalibration
NoAction
UseCalibrationValues

一般通过软指令触发光源开关：

```
success = m_framework.RunJob(command, LightControlAction.AutoOnAndOff, true, runJobParams);
```

在飞拍或者 3D 相机等需要硬件触发的：

```
command.WaitForDisplayResultCompleted = true;
success = m_framework.RunJob(command, LightControlAction.NoAction, true, runJobParams);
```


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Machine Support： 文件介绍

.vs	2023/4/21 12:50	文件夹	
ShelfAlignmentMachineSupport	2023/4/21 13:18	文件夹	
ShelfAlignmentMachineSupport.sln	2018/6/6 11:24	Microsoft Visual...	1 KB
ShelfAlignmentMachineSupport.suo	2021/12/20 15:30	Visual Studio Sol...	5 KB
ShelfAlignmentMachineSupport.v1...	2023/4/21 12:51	Visual Studio Sol...	88 KB
ShelfAlignmentMachineSupport.v1...	2021/12/20 15:49	Visual Studio Sol...	73 KB

- SimulatorCommandStrings
  - SetupCustomGUI()
  - Shutdown()
  - SimulatorCommandParserIDs
  - SimulatorCommandStrings
  - SimulatorPLCCommands
  - SingedNormDegrees(double degrees)
  - SortImages(int nTBIndex, string strRawImgPath, string strAnnotatedImgPath, int nErrCode)
  - str1AnnotatedImgPath
  - str1RawImgPath

打开 MachineSupport，找到“ SimulatorCommandStings” 进入

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Machine Support： 添加 command 指令

```
List<string> strings = new List<string>();
//strings.Add("P0");
strings.Add("P1,SN");
//strings.Add("P100");
strings.Add("P200,SN,0,0,0");
strings.Add("P300,SN,0,0,0,0.085,0.08");
strings.Add("P400,SN,0,0,0,0.085,0.08");
//strings.Add("P500");
strings.Add("CheckPosition,FOV");
strings.Add("CheckSize,FOV");
strings.Add("GROUP,SN,OK");
```

添加一个你想要的界面指令，可以先仿照上面的指令添加一个自己的指令，这个会在 MachineSupport 里重新生成后显示程序预设命令中

当然如果你不知道添加什么命令，可以先空着不添加，后面需要了再添加

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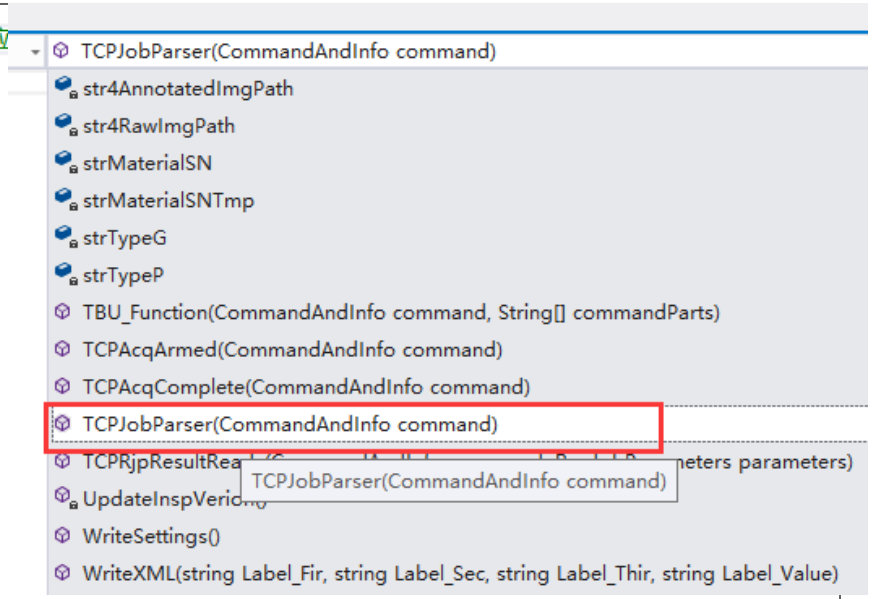
Machine Support： 添加 TCPJobParser

```
0 个引用
public string TCPJobParser(CommandAndInfo command) //收到客户端的指令，对指令进行解析，返回客户端的响应
{
    bool runStatus;
    string strResult = "";
    String strCmd = command.Command.Trim();
    string[] strArray = strCmd.Split(',');
    if (strArray.Length == 0) return "Unsupported Command";

    mStepName = strArray[0];
    if (command.IsSimulatorCommand)
    {
        mSN = strArray[1];
    }
    else
    {
        mSN = strArray[1];

    }
    try
    {
        switch (strArray[0])
        {
            case "P1":
                //Rest AlginTimes and AllExecTime
                mAlignTimes = 0;
                mAllExecTime = 0;
                if (mSN.Length < 3)
```



找到 TCPJobParser ，添加你上面命令对应的指令，这里分了 P1 和 P100 调用不同的 Calibration, 记得后面的 return SendData 中调用不同的函数，比如我们借调的 CamjobParserP1 和 CamjobParserP100

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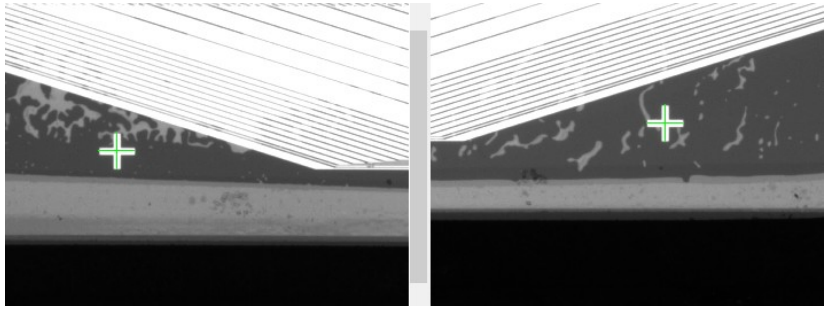
Machine Support： 添加 CamjobParserP1

```
public string CamjobParserP1(CommandAndInfo command, string[] strArray) // 方法: 这是一个处理联系ToolBlock的方法
{
    //Rest stopwatch
    spWatch.Reset();
    spWatch.Start();
    mStratTime = DateTime.Now.ToString("yyyyMMdd_HHmmss_fff");
    string reslut = "";
    bool runStatus;
    ulong partID = 0;
    RunJobParameters jobParamsP1 = new RunJobParameters(0, m_currentPartID);
    jobParamsP1.Inputs["I_RunNum"] = 1;
    jobParamsP1.Inputs["Str_SN"] = mSN;
    jobParamsP1.Inputs["Str_Command"] = command.Command.Trim();
    runStatus = m_framework.RunJob(command, LightControlAction.AutoOnAndOff, true, jobParamsP1);
    if (runStatus)
    {
        reslut = (string)jobParamsP1.Outputs["Str_SendData"];
    }
    else
    {
        reslut = (string)jobParamsP1.Outputs["Str_DefaultData"];
    }
    spWatch.Stop();
    mEndTime = DateTime.Now.ToString("yyyyMMdd_HHmmss_fff");
    mExecTime = spWatch.Milliseconds;
    mAllExecTime += mExecTime;
    Thread T1 = new Thread(new ParameterizedThreadStart(SaveImage));
    T1.Start(jobParamsP1);
    return reslut;
}
```

该方法通过抓取十字 mark 定位 CurObjectplacement



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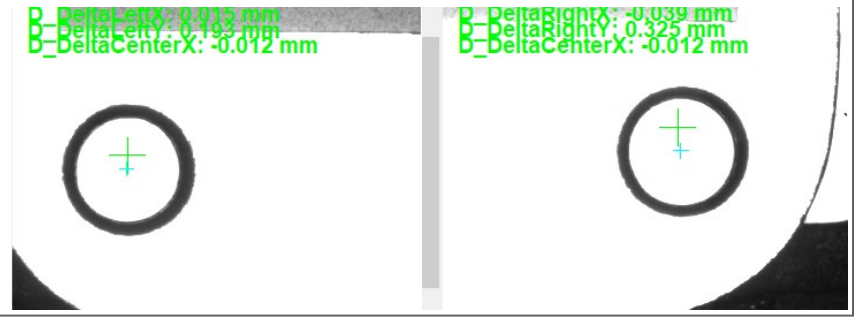
Machine Support： 添加 CamjobParserP100

```
public string CamjobParserP100(CommandAndInfo command, string[] strArray) // 方法: 这是一个处理联系ToolBlock的方法
{
    spWatch.Reset();
    spWatch.Start();
    mStratTime = DateTime.Now.ToString("yyyyMMdd_HHmmss_fff");
    string reslut = "";
    bool runStatus;
    ulong partID = 0;
    RunJobParameters jobParamsP100 = new RunJobParameters(0, m_currentPartID);
    jobParamsP100.Inputs["I_RunNum"] = 100;
    jobParamsP100.Inputs["Str_Command"] = command.Command.Trim();
    runStatus = m_framework.RunJob(command, LightControlAction.AutoOnAndOff, true, jobParamsP100);
    if (runStatus)
    {
        reslut = (string)jobParamsP100.Outputs["Str_SendData"];
    }
    else
    {
        reslut = (string)jobParamsP100.Outputs["Str_DefaultData"];
    }
    string dataStr = "";
    for (int i = 0; DataToController != null && i < DataToController.Count; ++i)
    {
        Tuple<string, double> entry = (Tuple<string, double>)DataToController[i];
        dataStr += string.Format(",{0:F3}", entry.Item2);
    }
    spWatch.Stop();
    mEndTime = DateTime.Now.ToString("yyyyMMdd_HHmmss_fff");
    mExecTime = spWatch.Milliseconds;
    mAllExecTime += mExecTime;
    return reslut;
}
```

该方法通过抓取圆圈 mark 定位 DesObjectplacement



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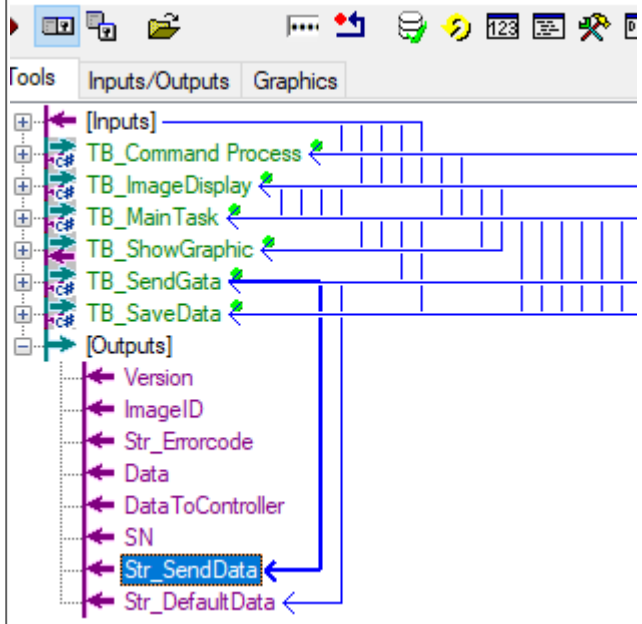
Machine Support： 和 VPP 进行数据交换和传递

比如我们在 Output 中添加 string 变量 Str_SendData ，将 X Y A 输入给它，当然你也可以添加或者减少数据



将 Output 中的 str_SendData 传输给 MachineSupport 中 strReturn 并返回，需要修改协议时也可以添加其他数据或者 ErrorCode 之类

```
    strReturn += (string)InspRunParams_TBU[j].Outputs["strSend"];
}

        }
        return strReturn;
    }
```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Machine Support： 和 VPP 进行数据交换和传递

将数据输入到 vpp 的 Input 中

eSupport.cs HACTemplate_MachineSupport

eSupport.HACTemplate_MachineSupport — Function_FinalInspection(CommandAndInfo command, int nDispla

```
CogTransform2DLinear MotionB_Pose = new CogTransform2DLinear();
MotionB_Pose.TranslationX = double.Parse(commandParts[i + 1])
MotionB_Pose.TranslationY = double.Parse(commandParts[i + 2])
MotionB_Pose.Rotation = CogMisc.DegToRad(double.Parse(command
m_Cave_Pose_Info[j].Camera_Pose = MotionB_Pose;
m_Cave_Pose_Info[j].CaveID = Cave_ID;
}

for (int i = 0; i < Cave_Num; i++)
{
    FrameworkConfiguration.gOnly.SetInspectionInputLink(GetCali
    RobotPose Moving_Pose = new RobotPose(m_Cave_Pose_Info[i].C
    // RobotPose Moving_Pose = new RobotPose(m_Cave_Pose_Info[i
    InspRunParams_Temp_Check[i] = new RunJobParameters(GetInspe
    InspRunParams_Temp_Check[i].Inputs["Input_Command"] = comma
    InspRunParams_Temp_Check[i].Inputs["CaveID"] = m_Cave_Pose_

    InspRunParams_Temp_Check[i].Inputs["MovingPoseX"] = m_Cave_
    InspRunParams_Temp_Check[i].Inputs["MovingPoseY"] = m_Cave_
    InspRunParams_Temp_Check[i].Inputs["MovingPoseDeg"] = CogMi

    InspRunParams_Temp_Check[i].Inputs["Mean"] = 255;

    if ((m_TerminalDatas_T2[m_Cave_Pose_Info[i].CaveID - 1] !=
    {
        InspRunParams_Temp_Check[i].Inputs["TerminalDatas_T2"]
        InspRunParams_Temp_Check[i].Inputs["TerminalDatas_T3"]
```

VisionTool Configuration - Inspection 5 - Insp 5 T4C Final Recheck

工具 | 输入/输出 | 图形

输入

名称	类型	值
MovingPoseX	System.Double	450.8
MovingPoseY	System.Double	58
MovingPoseDeg	System.Double	0
Input_Command	System.String	T4C,1,2018060...
AutoCom	System.Boolean	False
InputImage0	Cognex.VisionPro.CogImage8Grey	
InputImage0Home2D	Cognex.VisionPro.CogImage8Grey	Cognex.VisionP...
ProductName	System.String	Product 1
InputImageCameraSettings	System.String	Cam4_Exposur...
InputImage0CameraSettings	System.String	Cam4_Exposur...
InputImageLightChannels	System.String	
Mean	System.Double	255

输出

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作业

1. 用自己添加的 MachineSupport ，将第 2 个 Inspection 的 Output 端输出 X Y （x y 都是 double 类型）传到第 3 个 Inspection 中的 Input 中对应坐标 X Y

2. 假定调用 2 个实体相机，相机分辨率是 2448*2048 ，要求完成

a- 相机 1 实时界面时显示一条横线和一条竖线（标记视野中心，线显示绿色）

b- 相机 2 实时界面时显示三条横线和四条竖线（横线显示红色，竖线显示绿色）

使用合适的函数，完成上述要求

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Thank you

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